03 Technical requirements

TRD v0.5 Reference design and implementation contracts

v0.5 Updated 11 October 2026 Owner decision selects A and B with streaming for A

TRD v0.5 is the authoritative design and technical contract from 11 October 2026. It preserves v0.4 ingestion, retrieval, eligibility and safety contracts and adopts the owner-selected A/B answer methods in section 14. The approved owner manual governs motorcycle facts. Exact source approval scopes, applicability, usage rights and release gates remain binding; a methodology decision is not a public deployment approval.

1 Design philosophy and trust boundaries

This specification describes a version-locked, layout-aware multimodal RAG pipeline using only the supplied Royal Enfield New Himalayan 450 owner's manual with diagnostic session state. The LLM may translate a customer report into structured observations and paraphrase manual-backed instructions. No model may author a new repair step absent an applicable evidence object.

Layer	Authority	Failure response
Manual registry	Model/variant/version/checksum	Block when unverified or inconsistent
PDF parser + evidence store	Extracted source with page and bounding-box traceability	Flag uncertain/missing extraction; manual review needed
Dense/sparse retriever	Candidate relevance, not factual truth	Broaden search once; do not equate empty top-k to true absence
Document graph	Explicit, verified source dependencies	Ignore unverified semantic edges for technical instructions
Evidence sufficiency gate	Applicability, prerequisites, coverage and warnings	Clarify or abstain
Generative model	Natural-language presentation	A exact reviewed selection checks; B semantic claim verification; fail-closed delivery
State machine	Next eligible, documented diagnostic branch	OEM referral if no valid path remains

2 Versioned hosted ingestion pipeline

1. Register manual: OEM, model, year range, variant, document edition, original URL, local hash SHA-256, language and rights status
2. Parse each page with Docling or another layout-aware parser; retain PDF index, printed page label, headings, reading order, tables, images, warning/caution objects and bbox
3. Create typed evidence objects: PROCEDURE, STEP, CONDITION, WARNING, SPEC_TABLE, TABLE_ROW, FIGURE, SECTION, CROSS_REFERENCE
4. Apply hierarchical or hybrid chunking to the typed structure; enrich each child with section path and variant; preserve full parent procedure for answer assembly
5. For tables, repeat headers, dimensions/units and variant tags; never embed anonymous numeric cells without headers
6. For images, store page crop, caption, nearby explanatory paragraphs and figure relationship; uncertain diagram interpretation is a review flag
7. Build a dense vector index and sparse BM25 index; link both to stable evidence IDs; do not regenerate on every request
8. Build explicit edges from headings, figure links, in-document cross-references, preconditions, warnings and NEXT_STEP references

9. Run ingestion QA: page/object counts, missing pages, table header preservation, warning detection and extraction confidence flags

3 Core data contracts

Entity	Required fields
Manual	manual_id, oem, model, variant, year_range, edition, language, checksum, rights_status, source_uri
EvidenceObject	evidence_id, manual_id, type, text_or_asset_ref, section_path, pdf_page_index, printed_page_label, bbox, extraction_status
Chunk	chunk_id, evidence_ids, parent_procedure_id, index_text, dense_vector_ref, sparse_ref, variant_tags, token_count
DocumentEdge	from_evidence_id, to_evidence_id, edge_type, provenance, verified_boolean
DiagnosticSession	session_id, manual_id, bike_variant, current_symptoms, known_observations, attempted_steps, excluded_branches, status, version
StepAttempt	step_id, condition_id, outcome_enum, observed_details, supporting_user_turn_id, timestamp, superseded_by
EvidenceBundle	query_id, manual_checksum, evidence_ids, required_evidence_ids, attached_evidence_ids, missing_evidence_ids, dependency_manifest_version, dependency_review_status, conflicts, missing_conditions, applicability_status, sufficiency_status
Response	type, natural_language, claim_evidence_map, cited_pages, suggested_step_id, clarification_question, escalation_uri, trace_id
StepDefinition	step_id, manual_checksum, action_evidence_ids, parent_procedure_id, symptom_tags, applicability_rule_ids, required_evidence_ids, dependency_manifest_version, review_status, rule_version
ApplicabilityRule	rule_id, predicate_type, fact_key, operator, expected_value, unit, origin, source_evidence_ids, policy_version, clarification_evidence_ids, review_status, rule_version
EligibilityDecision	step_id, eligibility_status, condition_results, reason_codes, supporting_evidence_ids, rule_version, dependency_manifest_version, session_version
RetrievalDecision	policy_version/config, calibration_status, mode, uncertainty_signals, decisions, recovery_attempts, rerank_attempts, requested/fetched/missing_evidence_ids, pair_count, skip_reason, stages_seconds, final_action

4 Retrieval + evidence assembly contract

First-stage dense retrieval uses cosine similarity or the embedding model's recommended metric. Sparse BM25 retrieval handles exact warning labels, component codes and units. Fuse ranks with Reciprocal Rank Fusion, assemble initial parent and verified dependency context, and inspect ranking uncertainty separately from source applicability and evidence coverage. Conditionally rerank a bounded candidate pool before reassembly and the controller/answer gates.

The hosted prototype uses a fused pool of 20 candidates and selects five hits for assembly. One recovery may widen the fused pool to 40 and fetch exact reviewed dependency IDs. Reranking scores at most 20 candidates and 100 token-bounded query/passage pairs. At most 256 mandatory dependency IDs may be attached; an over-limit manifest fails as SYSTEM_ERROR and must not be silently truncated.

These limits and trigger thresholds are versioned starting hypotheses, to be calibrated on independently reviewed cases.

- Apply hard filters by selected manual ID, model/variant, publication version and language before ranking
- Edge expansion should be selective: always include required warnings/prerequisites; retrieve only referenced figures/specifications, not the whole manual
- For each candidate, compute evidence coverage over requested step, conditions, warnings, variant-specific values and referenced sections
- If an approved dependency set is incomplete, permit one bounded recovery that fetches the exact required objects and widens retrieval once. If dependencies remain unavailable, withhold technical advice and record a non-rider-resolvable source gap. Allow at most one subsequent reranking attempt when ordering remains uncertain and all source/applicability checks permit it; never loop between search, reranking and generation.
- Cosine similarity and reranker scores are relevance heuristics, never proof that an instruction is supported

4 1 Evidence dependency manifest

During versioned ingestion, bind each searchable child chunk to its full parent procedure or section and build a reviewed dependency manifest for every actionable step. The manifest identifies the action spans, applicability conditions, prerequisites, warnings, table headers and units, variant labels, and referenced figures or sections required to present that step. A section parent remains contextual evidence until its procedure and step boundaries have been reviewed.

The assembler returns required_evidence_ids, attached_evidence_ids and missing_evidence_ids for the selected candidate, together with dependency_manifest_version and dependency_review_status. missing_evidence_ids is the required set minus the attached set. Coverage is complete only when the dependency manifest is approved for the exact manual checksum, the action evidence is present, all required objects are attached, and their content and provenance pass extraction checks. An empty or unreviewed required set cannot establish sufficiency. Finding a table object is insufficient if its required headers, units or variant context are missing.

Perform the existing bounded recovery search for missing dependencies and record its outcome. A source gap that the rider cannot resolve must block the instruction and enter source review or abstention; do not ask the rider to supply a missing manual warning or table header. Retrieval outages, invalid index objects and other operational failures return SYSTEM_ERROR. Cross-encoder scores rank relevance and cannot approve a dependency manifest, infer a diagnostic order or waive missing evidence.

4 2 Conditional reranking and bounded recovery

The normal request path uses conditional reranking. Clearly ordered candidates with complete reviewed dependencies continue to the eligibility and answer gates. Reranking handles uncertain ordering among plausible candidates; evidence recovery handles required objects absent from the initial bundle. Similarity, RRF and cross-encoder scores cannot prove support or approve a procedure.

Trigger signals compare distinct parent sections: disagreement between dense and BM25 top sections, or a narrow relative RRF margin between the two leading sections. Sibling chunks from the same section do not create ambiguity. Zero BM25 matches provide no sparse-agreement signal. Initial experimental values are top-three section overlap below 0.5 or relative RRF margin at most 0.05. Mark these values PROVISIONAL_UNCALIBRATED; use reviewed tuning cases to calibrate them and held-out cases to assess their effect before a release decision.

Apply the routing policy in order. A known safety stop retains its verified warning and skips model work; a known system failure remains SYSTEM_ERROR; a known unsupported request abstains. Missing, stale or unapproved source/dependency review enters source review or abstention. Unconfirmed applicability enters one focused clarification; a confirmed mismatch blocks the action. An approved missing

dependency triggers one exact-ID fetch and bounded retrieval expansion. Complete eligible evidence with uncertain ordering triggers one cross-encoder attempt; otherwise continue through the remaining gates.

Reassemble after reranking and recheck the dependency set. Retain mandatory action, warning, prerequisite, table/header/unit and figure objects for the selected reviewed step. Reranking does not select a diagnostic step or infer OEM order. The controller selects an eligible step; final claim validation follows the A or B method in sections 6 and 14.

Operational failures in embedding, retrieval, assembly or the cross-encoder yield SYSTEM_ERROR with a typed error class and recorded attempts. A query exceeding the current cross-encoder's 128-token input bound, or work exceeding 100 query/passage pairs, skips that inference with an explicit reason and keeps ranking unresolved. Preserve the full query and source text; never silently truncate a rider report or mandatory evidence. A source gap cannot be resolved by asking the rider to supply a missing manual warning or table header.

Return a RetrievalDecision with policy version/configuration, calibration status, execution mode, uncertainty signals, decision history, recovery/rerank attempt counts, exact missing/fetched evidence IDs, pair count, skip reason and stage durations. Conditional mode is the normal code path; off and always_diagnostic modes exist only for ablations. Always-diagnostic execution can bypass the scheduling decision but cannot bypass evidence, applicability, safety or answer-release gates. A generic insufficient-bundle flag must never trigger reranking by itself.

Candidate bundles without approved applicability and dependency scopes remain blocked. Owner-delegated coordinator approval now exists for 161 exact source-content/initial-response scopes and the selected Himalayan BS6 manual scope in release_manifest_v3.json. Consume its exact bindings; do not extend approval to changed objects, full adaptive transitions, general vision or deployment. Conditional retrieval scheduling remains versioned and reuses unchanged document embeddings.

Evaluate off, always and conditional policies on the same independently reviewed cases. Record complete evidence recall, rescued and regressed cases, proper abstention, wrong-variant leakage, trigger precision/recall, reranking frequency, per-stage latency, TTFO and cost per correctly handled case. Adopt calibrated conditional reranking for release only if it improves reviewed outcomes without violating latency/cost and safety gates. The existing 27 draft in-sample page-coverage probes cannot establish that result.

Cross-encoder retrieval reference:
https://www.sbert.net/examples/sentence_transformer/applications/retrieve_rerank/README.html

5 Diagnostic state machine and adaptive branching

A deterministic controller stores the diagnostic session separately from the LLM chat transcript. The controller accepts typed feedback and selects eligible actions only from the verified procedure graph. Generation occurs after next-step selection, never before.

5 1 Reviewed step registry and rule authoring

Create candidate StepDefinition and ApplicabilityRule records from the original owner manual during ingestion. A layout parser or model may propose the records, but an independent reviewer must check each action, condition, prerequisite, warning and stopping condition against the original pages before it becomes runtime-authorized. Record the reviewer, decision, source spans and registry version. Draft, rejected, stale or checksum-mismatched rules must not authorize an action or a cached answer.

Every manual-derived predicate must retain its supporting evidence IDs and page provenance. Record origin as MANUAL for OEM conditions and PRODUCT_POLICY for application behavior such as avoiding repeated attempts; policy rules carry their own version and must never be presented as OEM instructions. Relevance tags and rider intent mappings help select candidates, while the reviewed predicates establish applicability. Uncertain interpretation of the rider report remains an unknown observation until clarified.

When the manual lists possible causes or independently documented checks without branch conditions or an explicit next-step order, preserve those limits. Do not infer a condition, eliminate an undocumented branch, diagnose a fault from a negative clue or claim an OEM ranking from retrieval scores. The existing SM-04 policy governs selecting among independently documented eligible checks.

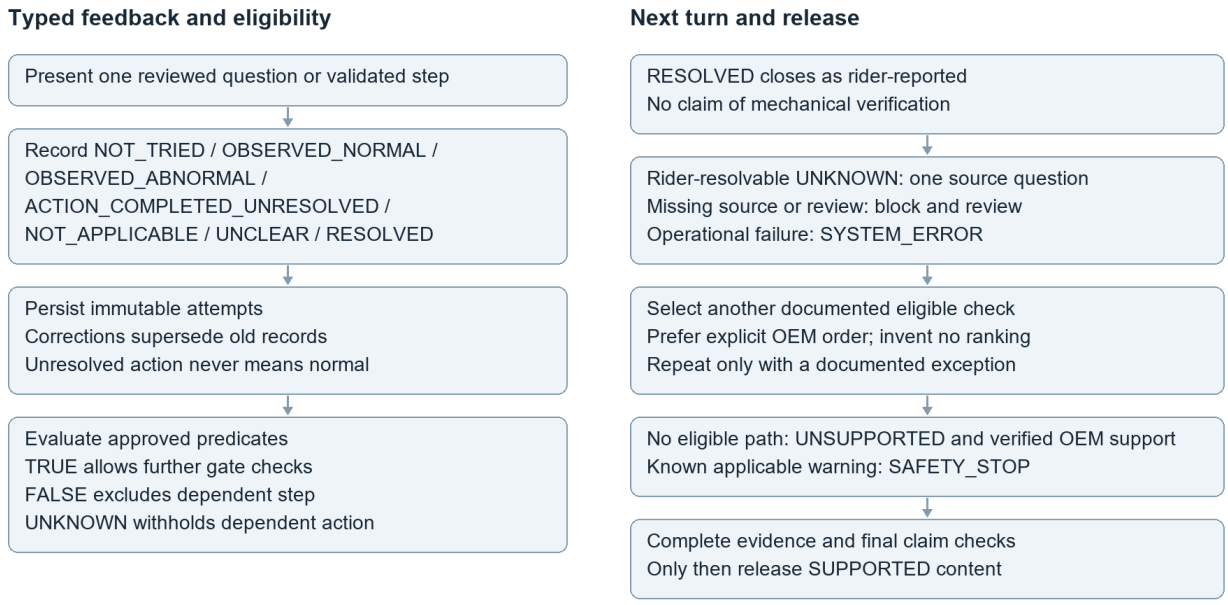
5 2 Three valued eligibility checks

Evaluate reviewed conditions against typed session facts using TRUE, FALSE and UNKNOWN. A condition is UNKNOWN when its observation is missing, uncertain or invalid; a confirmed mismatch is FALSE. A step is ineligible when any required condition is FALSE, its source or review status is invalid, its branch has been validly excluded, or its attempt history blocks repetition. If no required condition is false but a relevant condition is unknown, withhold the action and ask one reviewed source-derived question when the rider can reliably resolve it. Unknown is never treated as true or normal.

An action becomes eligible only when the manual and variant match, the rule is approved, the symptom or documented branch applies, required conditions and prerequisites are true, no applicable safety rule prohibits it, and the attempt history allows it. Evaluate the evidence-completeness and answer-release gates separately before issuing technical text. A direct factual lookup uses the same applicability and evidence gates without fabricating a diagnostic step.

Implement rule predicates with allowlisted typed comparisons such as enum equality, boolean facts and numeric comparisons with explicit units. Treat the PDF and model output as data; do not execute extracted expressions as code. Return an EligibilityDecision with each condition result, reason code, evidence reference, registry version and session version. ACTION_COMPLETED_UNRESOLVED records an attempted action and never supplies an OBSERVED_NORMAL fact. Optional interpretation models may propose typed observations or rank already eligible candidates; their output cannot bypass the controller gates.

5 3 Controller behavior



Rule ID	Controller behavior
SM-01	Each suggested diagnostic step gets a stable step_id and an immutable attempt record; updates supersede earlier observations rather than delete them
SM-02	Only structured observed outcomes can eliminate a branch; “tried but still failing” must not be converted into “normal”
SM-03	Next-step eligibility uses approved StepDefinition and ApplicabilityRule records for the selected checksum/variant, typed condition results, satisfied prerequisites, no prohibiting safety condition and permitted attempt history. Relevance alone cannot authorize a step; complete evidence is required separately before release.
SM-04	Prefer explicit OEM NEXT_STEP order; when no explicit order exists, only recommend another independently documented eligible check and do not imply OEM prioritization
SM-05	When a reviewed branch condition is UNKNOWN, withhold the dependent action and ask one source-derived question if the rider can resolve it. If a source or extraction gap cannot be resolved by rider feedback, route to review/abstention; operational failures remain SYSTEM_ERROR.
SM-06	Do not repeat attempted steps unless the customer corrects an observation, new symptoms change applicability, or the manual explicitly requires a repeat; explain why
SM-07	If no eligible documented step remains, issue OEM escalation with a concise traceable recap
SM-08	Model/version switch invalidates procedural branch assumptions and requires explicit reconfirmation
SM-09	Safety warnings take precedence over ordinary branch selection; unsupported safe workarounds are forbidden
SM-10	When a user reports resolution, close as customer-reported, not mechanically verified

6 Evidence sufficiency and output validation

Apply the deterministic rules gate to every method: valid manual and variant, complete approved dependencies, satisfied prerequisites, required warnings, no explicit conflict and valid immutable citations. In A, the model selects indices from exact reviewed wording and the server validates the selection and releases only the bound wording; no free technical paraphrase or separate online semantic verifier is permitted in A. In B, the model composes a concise draft and a separate semantic support evaluator checks each proposed claim against its source spans before release. Reject output that fails the applicable method. A receives a post facto semantic verification audit before UI integration, as specified in section 14.

6 1 Response routing and release

The controller selects the response route from structured eligibility decisions, the evidence bundle and gate results. A known applicable stop condition yields SAFETY_STOP and retains its verified warning even if other processing fails. Otherwise, a required component failure yields SYSTEM_ERROR. A resolvable missing applicability fact or documented branch observation yields AMBIGUOUS and one focused question. After bounded recovery, an unsupported request, an unresolved source gap or exhausted documented paths yields UNSUPPORTED with the verified support route.

SUPPORTED requires an eligible action or applicable factual answer, complete approved dependencies, valid citations, satisfied safety and prerequisite checks and successful method-specific claim validation. A requires exact reviewed wording and valid allowlisted selections; B requires deterministic checks plus successful semantic verification. Both release only after validation and durable session save. Rejected drafts remain withheld. Bounded operational generation/verification failures return SYSTEM_ERROR; established source gaps yield UNSUPPORTED. A post facto audit does not retroactively authorize unsafe output.

Gate result	Behavior
SUPPORTED	Release only after applicability, complete dependencies, warnings, citations and method-specific claim validation under A or B pass.
AMBIGUOUS	Ask one focused source-derived question about an unknown branch observation or resolvable applicability fact; withhold the dependent action.
UNSUPPORTED	Abstain with “could not verify an applicable procedure” and verified OEM contact option
SAFETY_STOP	Display the applicable source warning, stop unsafe action and refer to authorized service as supported
SYSTEM_ERROR	Explain temporary inability to verify; do not claim absence from the manual

Even a high-scoring semantic verifier can make mistakes; guarantee only that the application follows the fail-closed protocol, not that all semantic classifications are infallible. Output citations must map to immutable evidence object IDs, not model-created page numbers.

7 Multimodal path

- Photo input: validate size/type, run vision extraction into observations {component_hint, visible_feature, uncertainty, needs_clarification}; no root-cause diagnosis from vision alone
- Document diagrams: associate image with figure, heading, parent procedure and original page crop; require model verification against the image for image-dependent claims
- Voice input P1: use speech-to-text (Sarvam or alternative), show/edit transcript, and route to identical RAG evidence checks; speech duration, STT latency and STT spend are separately reported
- No permanent public-demo storage of image/audio by default; avoid putting raw media or personal text into traces

8 Service interfaces (provisional)

Contract	Core parameters	Result
POST /sessions	manual_id, bike_variant, optional year	session_id, manual metadata, status
POST /sessions/{id}/turns	text; optional image; optional confirmed transcript; previous typed outcome	SUPPORTED / AMBIGUOUS / UNSUPPORTED / SAFETY_STOP / SYSTEM_ERROR
POST /sessions/{id}/feedback	step_id, outcome_enum, observed_details	updated session version, next-turn eligibility
GET /evidence/{id}	source evidence ID	Approved paraphrase and immutable citation metadata only; no end-user PDF, excerpt or crop under current display policy
GET /health	dependency checks	app/index/model connectivity status

For a small Streamlit deployment these may remain internal Python service boundaries rather than external REST endpoints. Keep request/response contracts stable so a future FastAPI backend can be introduced.

9 Latency, caching and cost engineering

Technique	Benefit	Guardrail
Prebuilt indices + graph	Avoid online parsing and embedding of manual	Index keyed by immutable document checksum
Selected-manual session cache	Avoid repeated manual selection and variant lookups	Invalidate on variant change
Exact reviewed procedure path cache	Avoid repeated generation for approved exact paths	Key checksum, confirmed variant, intent/symptoms, typed outcomes, flow/controller, policy and prompt versions; invalidate on relevant rule/review/dependency changes; draft/stale entries miss
Small evidence bundle	Fewer LLM input tokens	Mandatory warnings/conditions are never dropped
Conditional reranker	One attempt for uncertain candidate ordering after source and coverage checks	Versioned provisional thresholds; reviewed calibration; bounded pool; mandatory evidence retained
Bounded graph expansion	Avoid token and network explosion	Whitelist edge types and cap hops
Optional streaming STT	Better perceived voice responsiveness	Do not emit answer from unstable partial transcript

Initial benchmarks: text warm p50 <= 2.5 s and p95 <= 5 s; photo p95 <= 10 s; log cold start separately. Measure end-to-end and each stage with p50/p95 and cold/warm split. Provider pricing must be captured at deployment time rather than hardcoded. Measure time to first output (TTFO) from confirmed rider input to the first releasable answer token. A loading indicator is excluded. Technical content is withheld until the required claim and citation checks pass. Report fast-path and recovery/reranking p50/p95, their combined p95, fallback frequency and cold starts separately; reduced average latency alone does not establish a lower p95.

10 Observability design

Event / span	Fields to capture
turn_received	trace_id, session_hash, modality, manual_checksum, input_size_bucket
visual_interpretation	visible_obs_count, uncertainty, duration_ms, model_version
retrieval_dense / retrieval_sparse	top_candidate_ids, ranks/scores, duration_ms, index_version
rerank / graph_expand	policy_version, calibration_status, mode, trigger_reasons, ranked_ids, pair_count, skip_reason, verified_edges, added_dependencies, attempt_counts, duration_ms
sufficiency_gate	decision, failed_gate_codes, missing_evidence_ids, duration_ms
session_transition	step_id, previous_outcome, current_outcome, excluded_branches, state_version
generation / citation_check	input/output tokens, model_version, claim_support_results, duration_ms
turn_completed	final_status, end_to_end_ms, estimated_cost, feedback, error_class

Event / span	Fields to capture
step_eligibility	candidate_step_ids, condition_results, reason_codes, missing_evidence_ids, rule_version, dependency_manifest_version, session_version, duration_ms

- Use OpenTelemetry-compatible spans and a tracing/evaluation backend such as Langfuse. Scrub raw user media and redact accidental personal information.
- Report weekly/experiment aggregates: unsupported-claim incidence, evidence recall, proper abstention, answerable coverage, repeated-step rate, p95 latency, per-stage failures and cost per correctly handled case

11 Deployment, security and reliability

- Host a Streamlit UI at a shareable URL; keep API keys server-side in secrets management; pin dependencies and models
- Package a small prebuilt index snapshot and verified document graph or use managed Qdrant if local resource limits are unacceptable
- Apply file size/type limits, per-user rate limits and a per-day spending cap; sanitize PDF text as data, not executable instructions
- Do not include proprietary manuals in an unrestricted public repository without verifying usage rights; use private asset storage or a permitted manual
- Graceful errors use SUPPORTED, AMBIGUOUS, UNSUPPORTED, SAFETY_STOP and SYSTEM_ERROR under section 6 1. Review gaps block advice; rider-resolvable applicability facts clarify; confirmed mismatches block; operational failures remain SYSTEM_ERROR.

12 Evaluation, ablations and versioning

Evaluate at least 25 gold records for an MVP; expand to 50–100 with the supplied manual. Include multi-turn transcripts and photo inputs, required evidence sets, forbidden claims and outcome expectations. Always annotate independently from the tested model where feasible.

12 1 Eligibility and assembly regression cases

Include independently reviewed cases that verify rule provenance and expected eligibility decisions, in addition to retrieval relevance and answer wording. Record expected condition results, exclusion reasons, required evidence IDs, response route and allowed next step for each case.

- TRUE, FALSE and UNKNOWN facts select, exclude or clarify the documented branch correctly; unknown facts and unresolved attempts never become normal observations.
- Draft, stale, wrong-variant and checksum-mismatched step rules cannot authorize technical text or cache hits; every manual-derived predicate resolves to its reviewed source spans.
- A relevant child chunk with a missing prerequisite, warning, table header/unit or referenced figure cannot pass the completeness gate. A missing or unapproved dependency manifest also fails closed.
- Distinguish a resolvable observation from a source gap, a known safety stop and an operational failure. Exhausted paths escalate; relevance scores cannot invent a step or repeat a completed action without the documented exception.

Keep the existing release targets. Report unsupported eligibility approvals and unjustified repeat steps alongside next-step correctness, complete evidence recall, answerable coverage and proper abstention; adjudicate failures against the original manual and reviewed policy rules.

Experiment	Independent variable	Measures
E0	Dense-only vs dense+BM25	Recall@5, complete evidence recall, latency

Experiment	Independent variable	Measures
E1	Flat chunks vs typed hierarchical/parent-child	Procedure completeness, warnings retained
E2	No graph vs 1-hop/2-hop verified expansion	Multi-section evidence recall, cost, false joins
E3	Off vs always vs conditional cross-encoder; optional late-interaction alternative	Complete evidence recall, rescues/regressions, trigger accuracy, fallback rate, TTFO p95, cost
E4	No verifier vs deterministic+semantic support gate	Unsupported answer risk vs answerable coverage
E5	Text vs photo / speech input	Observation accuracy, grounding and total latency

Version together: manual checksum, parser version, chunking parameters, embedding model, index build, graph build, step registry, applicability rules, dependency manifests and review decisions, controller policy, retrieval policy and trigger thresholds, calibration dataset and approval, reranker, generator, system prompt, verifier and evaluation set. Rule, review, dependency or policy changes invalidate affected exact-cache entries and require revalidation of affected cases. All comparisons must use the same held-out cases.

13 Unresolved engineering risks

- Visual table or diagram extraction could silently lose prerequisites, units or warnings
- Manual may not contain a true decision tree or enough alternatives for adaptive troubleshooting; OEM referral remains valid end state
- Closed-model semantic validators may exhibit false acceptances and false refusals; measure both
- Hosted free-tier cold starts and egress constraints can dominate model-side latency
- Model identity hallucination from photos can cause incorrect manual routing; require confirmed bike variant

Supporting vendor/research documentation is listed in the companion To Build document.

Recorded implementation status

Recorded implementation: hosted BGE M3 candidate index index_a3282c4d98cf54595e4ba168 has 532 children and 277 section parents; authenticated Cloud Run Qdrant restores immutable private vectors on cold start. The historical backend smoke passed 11/11. Isolated A/B/C experiments and verified delivery hooks are measured; the A/B production service and browser/UI delivery boundary require integration and separate validation.

The exact 161 source-content/initial-response scopes have coordinator approval under owner delegation in release_manifest_v3.json, checksum b8f8c7b5a9e1a299f0a90eecbf10127779b62b35fc105f0cd587de433e235298. The 17-test approval consumer validates 187 bound files. Remaining limitations include full adaptive transitions, general vision, changed runtime objects and source-specific gaps outside those scopes. End-user PDF, excerpt and figure delivery remain prohibited by source_manifest.json; approved paraphrases and citation metadata are the UI content boundary.

Authorized repository/project: Maneshwar-Kumar/motomedic-ai and motomedic-ai-20261009; US$10 project scope. Non-recurring INR800 budget starts 9 October 2026; alerts INR400/640/800 and simulation-checked live billing stop INR640 with retries. Reporting delays prevent an exact instantaneous cap. Do not re-enable stopped billing without owner authorization. Public UI daily/rate/upload limits remain pending.

Deployment remains the recorded v0.3 candidate boundary. Repository retrieval scheduling and isolated experiments do not establish a deployed rider service. This v0.5 amendment governs A/B and A streaming; consult section 14 and the deployment, smoke, source and v4 result records.

14 Owner selected answer methods and delivery

Decision recorded 11 October 2026. Retain A and B as the application methodologies. Use verified streaming delivery for A. Exclude C from the current UI path because it failed the clock case in both passes and took excessive paced time in the successful cases. This is the owner decision based on a bounded engineering comparison, not a general model quality or p95 finding.

Method	Selected behavior	Release requirement
A component selection	Select among exact reviewed wording; deliver by streaming	Validate exact indices, scope, dependencies and session; emit the complete approved unit after durable save
B short generation	Retain concise source-grounded composition with a separate verifier	Withhold the whole draft until semantic verification, deterministic gates and durable save pass
C verified independent blocks	Retain historical experiment only; do not wire into the current UI	No release route selected; no silent fallback from A/B to C

Streaming for A means HTTP/SSE delivery of a complete verified answer unit. Do not expose raw selection JSON, model token fragments or partial cautions as repair guidance. Preserve terminal complete/error events and session ownership. The UI distinguishes provider TTFT from first safe output; a progress indicator is not TTFO. B may use transport events after its whole-answer gate, but the owner has specifically selected A streaming.

14 1 Evidence and latency contract

Hosted run 38122505406 at commit e910c86d2fb8a0a2f955bbea4fa4cbedbe64d82d used gemma-4-26b-a4b-it with minimal thinking on GEN003 mirrors, EXP003 signal cancellation and GEN026 clock settings. Nine initial turns and nine immediate repeats produced 8/9 SUPPORTED outcomes in each pass. All six A/B initial cases passed. C clock delivered no output in either pass. The run made 27 attempts with zero retries: 26 known HTTP200 results and one unknown status.

Report retrieval wall time, provider TTFT, request-relative TTFT, actual first verified output and full completion separately. Keep fresh-worker startup, benchmark cooldown and idle provider-admission waits distinct. Removing idle waits arithmetically is not an observed unpaced run. Actual receipt was measured on hosted localhost HTTP/SSE; public network and browser rendering remain unmeasured. Three cases cannot establish p95 or architecture-wide superiority.

The candidate cache remains bounded and ephemeral. Similarity only locates a candidate; exact source, variant, selected step, typed state/history and runtime approval bindings authorize reuse. Only reviewed wording is stored, fresh retrieval and gates remain mandatory and B retains verification on hits. Three A repeats hit; no B/C repeats hit. Concurrent dense/BM25 retrieval remains implemented, but three paired controls demonstrated no latency gain.

14 2 Work before UI integration

First, run the removed semantic verifier post facto on the exact A outputs from all three cases, with repeat provenance retained. Bind the audited wording, claims, warnings and citations to the approved source and existing run. Record valid acceptance, valid rejection, invalid schema and provider/system failure separately; do not regenerate the answers or infer approval from a schema failure. This task is a new audit, not permission to alter old results or widen A beyond approved wording.

Second, investigate current Cloud Run minimum-instance pricing for the actual region, CPU, memory and billing mode, including idle cost, free tier and existing budget impact. Check current startup CPU boost settings and pricing and propose a controlled benchmark. Fresh-worker readiness was 56.64 seconds excluding controls, including 38.25 seconds for dense model/index load; this is not measured Cloud Run request cold latency. Do not change deployment, minimum instances, boost or billing as part of research.

The companion 03_TRD_Architecture_Decision_Note.docx records selected and rejected answer methods, measured retrieval/cache/streaming controls and broader design alternatives, distinguishing completed comparisons from options merely considered. Preserve the v4 raw results and independent audit as evidence. The concept, PRD and evaluation documents dated 9 October retain historical v0.4 language; this v0.5 amendment takes precedence for A/B method-specific validation and A streaming.